MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING
(A Unit of Rajalaxmi Education Trust®, Mangalore)
Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution

Machine Learning Operations (23AIPE312)

Module 1 & 2 Question Bank

Prepared by Manjunath Prasad H. R., Department of AIML, MITE

About this Document

This question bank has been prepared to help students prepare for the Continuous Internal Evaluation in Machine Learning Operations. The questions and answers are framed by the course faculty, strictly in line with the prescribed syllabus and the textbook *Introducing MLOps* (Treveil et al., O'Reilly, 2020). AI assistance was used for the formatting and layout of the document.

Scope

32 questions (16 per module, 8 of 8 marks and 8 of 6 marks), each mapped to a syllabus topic and a textbook page.

- **Syllabus:** 23AIPE312 Machine Learning Operations, Module 1 (Introduction to MLOps) and Module 2 (Process in Model Development and Versioning).
- **Textbook:** Treveil et al., *Introducing MLOps*, O'Reilly, 2020 — Chapter 1 (pp. 3–11), Chapter 3 (pp. 23–38), Chapter 4 (pp. 41–58).
- **Not covered:** Chapter 2, *People of MLOps* (pp. 13–22: Subject Matter Experts, Data Scientists, Data Engineers, Software Engineers, DevOps, Model Risk Manager/Auditor, ML Architect). Only its table of contents was available, so no questions were set from it.
- **Answer length guide:** an 8-mark answer expects about 8 distinct points or a point plus explanation per mark band; a 6-mark answer expects about 6.

Module 1 – 8 Mark Questions

Q1. Define MLOps. Explain the three key reasons why managing machine learning life cycles at scale is challenging. (8 marks)

Topic: Defining MLOps and Its Challenges · Ch. 1, pp. 4–5

Definition (2 marks): MLOps is the standardization and streamlining of machine learning life cycle management. It helps organizations generate long-term value and reduce the risk of data science, ML and AI initiatives. With decision automation (decisions made without human intervention), models become more critical and managing their risk becomes a top-level concern.

Three reasons (6 marks):

1. **Many dependencies.** Data changes constantly and business needs shift too. Results must be relayed back to the business continually, so that the model's behaviour on production data still matches expectations and still addresses the original goal.
2. **Not everyone speaks the same language.** The life cycle involves business, data science and IT teams. These groups use different tools and often lack shared fundamental skills, so there is no common baseline for communication.
3. **Data scientists are not software engineers.** Most specialize in model building and assessment, not in writing applications. Many juggle several roles and are stretched thin, which becomes serious at scale with more models to manage. Staff turnover makes it worse: data scientists end up managing models they did not create.

The simple life cycle (Fig. 1-2) understates this; the realistic picture (Fig. 1-3) involves many people with different skills and tools.

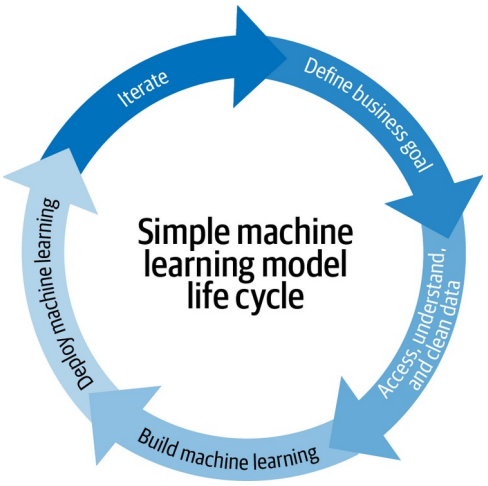


Figure 1-2. A simple representation of the machine learning model life cycle (Source: Treveil et al., Introducing MLOps, 2020)

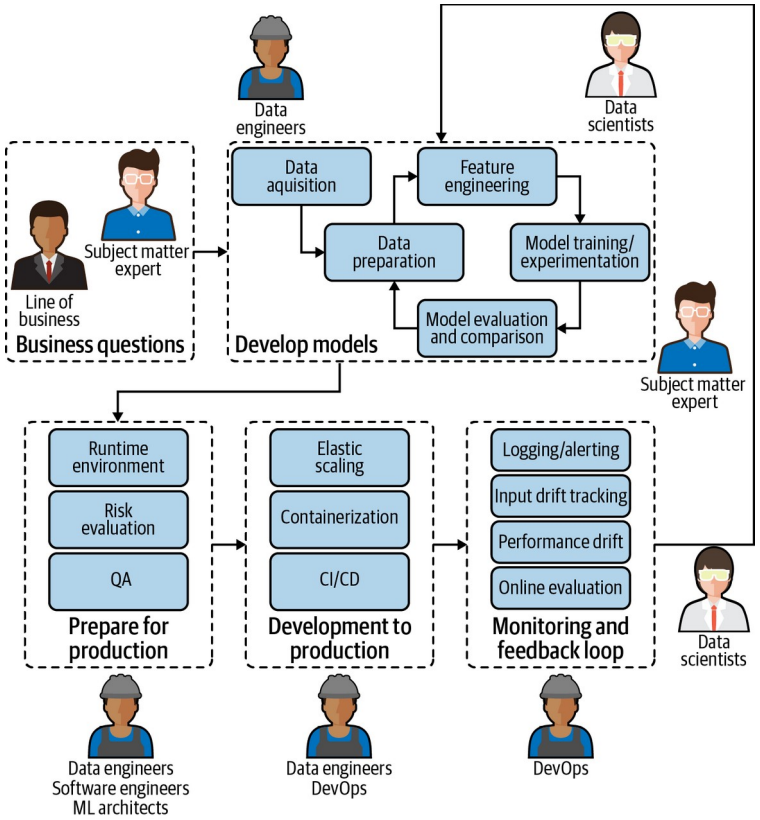


Figure 1-3. The realistic picture of a machine learning model life cycle inside an average organization (Source: Treveil et al., Introducing MLOps, 2020)

Q2. Compare MLOps with DevOps. How do DataOps, ModelOps and AIOps relate to MLOps? (8 marks)

Topic: Defining MLOps and Its Challenges · Ch. 1, pp. 3, 6–7

What MLOps and DevOps share (2 marks):

- Robust automation and trust between teams
- Collaboration and increased communication between teams
- The end-to-end service life cycle (build, test, release)
- Priority on continuous delivery and high quality

The critical difference (2 marks): Deploying software code is fundamentally different from deploying ML models. Software code is relatively static, but data is always changing, so models must keep learning and adapting to new inputs. Because an ML model is made of both code and data, MLOps is a new and unique discipline.

Term	Meaning	Relation to MLOps
ModelOps	Emerged with MLOps in 2018–2019; argued to be more general, covering any model (e.g., rule-based)	Largely used interchangeably; the textbook uses “MLOps” for the ML life cycle

Term	Meaning	Relation to MLOps
AIOps	Using AI to solve operational challenges (AI for DevOps), e.g., predictive maintenance for network failures	A different topic entirely, outside MLOps
DataOps	Introduced by IBM in 2014; delivers business-ready data quickly, focusing on data quality and metadata management	Intersects with MLOps, but MLOps adds more robustness through its key features

DataOps example (2 marks): If data a model relies on changes suddenly, a DataOps system alerts the business team to treat recent insights carefully and notifies the data team to investigate or roll back a library upgrade and rebuild the affected partition.

Why now (2 marks): Teams managed without centralized processes because they were not deploying models at large enough scale. They now need a multi-stage, multi-discipline, multi-phase process in a heterogeneous environment.

Q3. Explain how MLOps helps to mitigate risk, covering risk assessment and risk mitigation. (8 marks)

Topic: MLOps to Mitigate Risk · Ch. 1, pp. 7–9

MLOps matters to any team with even one model in production, because continuous performance monitoring and adjustment may be essential. MLOps has a cost, so a cost-benefit evaluation is done per use case.

Risk assessment (4 marks): Risks vary widely; a monthly marketing-offer recommender carries far less risk than a travel site whose pricing depends on a model. The analysis should cover the risk that:

1. The model is unavailable for a period of time
2. The model returns a bad prediction for a sample
3. Model accuracy or fairness decreases over time
4. The skills needed to maintain the model (data science talent) are lost

Risk is larger for models deployed widely and used outside the organization. Assessment uses two metrics, the **probability** and the **impact** of the adverse event; their combination gives the model's **severity** (Fig. 1-4). It is done at project start and reassessed periodically.

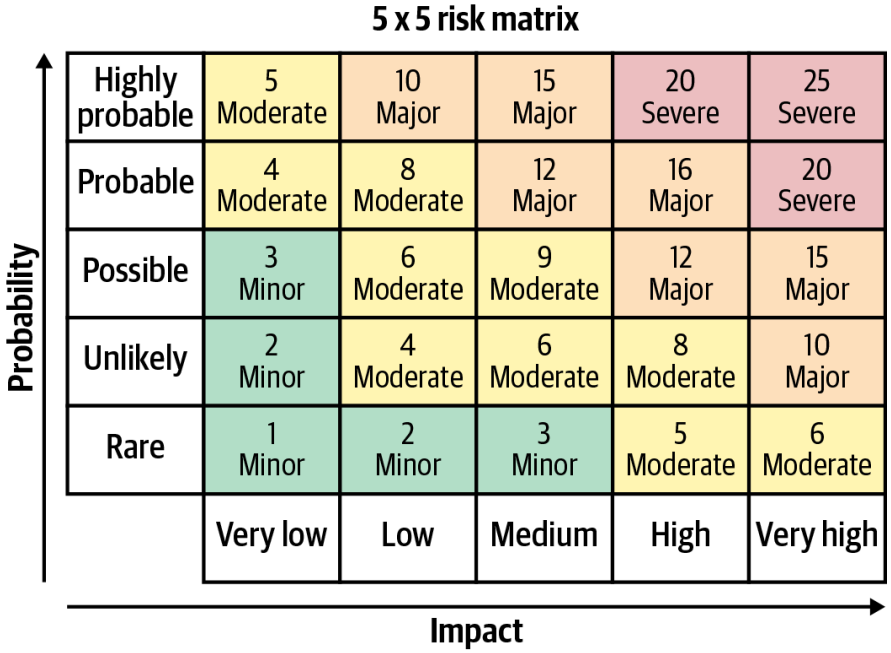


Figure 1-4. A table that helps decision makers with quantitative risk analysis (Source: Treveil et al., Introducing MLOps, 2020)

Risk mitigation (4 marks):

- MLOps becomes critical once a centralized team runs more than a handful of models; without standardization there is no global view of model states.
- A model's performance can often only be fully assessed in production, because training data must reflect production data. If production changes, performance can drop rapidly.
- Models are sensitive to their environment. They are built on open-source software (scikit-learn, Python, Linux), so production versions must match the versions the model was verified on.
- Deployment is the beginning, not the end: monitoring ensures expected behaviour and tells the organization how broadly each model is used.

Q4. Explain the stages of model development as a key feature of MLOps. (8 marks)

Topic: Model Development · Ch. 3, pp. 24–27

1. **Establishing business objectives.** Development starts with an objective, e.g., reducing fraudulent transactions below 0.1%. Objectives bring performance targets, infrastructure needs and cost constraints, captured as KPIs that later let production models be monitored. Objectives also include change management; e.g., the required transparency influences algorithm choice.
2. **Data sources and exploratory data analysis (EDA).** Subject matter experts and data scientists search for suitable input data, often the hardest part. EDA builds hypotheses, identifies cleaning needs and informs feature selection, visually or statistically.
3. **Feature engineering and selection.** Raw data is transformed into features (fixed-size arrays of numbers, the only object ML algorithms understand). It includes data cleansing, which can take the most project time.

4. **Training and evaluation.** Training is iterative: algorithms tested, features generated, selections adapted, hyperparameters tuned. It is the most compute-intensive step. An experiment tracking tool records data, features, parameters and metrics for side-by-side comparison. The best solution needs quantitative criteria (accuracy, error) and qualitative ones (explainability, ease of deployment).
5. **Reproducibility.** Significant model versions are saved with enough information about the environment to reproduce the same results from scratch. This needs version control of all assets, including data, and a record of the software environment; otherwise the model cannot be handed to DevOps reliably.
6. **Responsible AI.** DevOps must know how to verify the model; regulated industries document how it was built and tuned. Documentation (partly automated) and explainability techniques (partial dependence plots, subpopulation analyses, Shapley values, what-if analysis) address this.

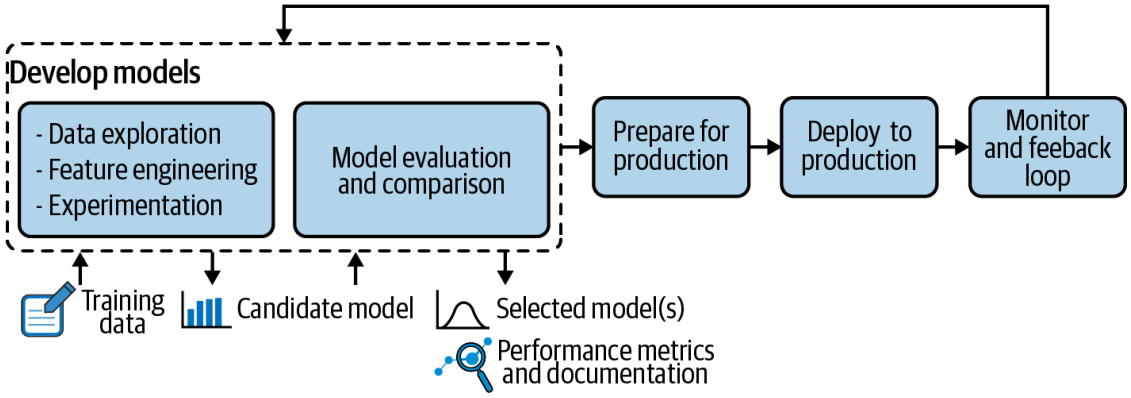


Figure 4-1. Model development highlighted in the larger context of the ML project life cycle (Source: Treveil et al., Introducing MLOps, 2020)

Q5. Explain productionalization and deployment of ML models: deployment types, model contents, containerization and deployment requirements. (8 marks)

Topic: Productionalization and Deployment · Ch. 3, pp. 27–29

Productionalization is the domain of software engineers and DevOps; poor collaboration with data scientists causes delays or failed deployments.

Deployment types (2 marks):

- **Model-as-a-service (live scoring):** the model sits in a simple framework exposing a REST API endpoint that answers requests in real time.
- **Embedded model:** the model is packaged into an application that is published, e.g., an application that batch-scores requests.

Model contents (1 mark): Code (Python, R or Java) plus data artifacts. Runtime and package versions must match production, or predictions may differ.

Portable formats (1 mark): PMML, PFA, ONNX or POJO reduce dependency on the production environment, but each supports limited algorithms and may behave subtly differently from the original.

Containerization (1 mark): Docker containers are lightweight alternatives to virtual machines, giving each model a self-contained environment. They enable blue-green deployment and elastic scaling; Kubernetes orchestrates many containers, in the cloud or on-premise.

Deployment requirements (3 marks): Rapid automated deployment is preferred. Short-lived self-service apps may use a single-step push to production, with resources capped by Linux cgroups and simple UIs in Flask. Customer-facing, mission-critical cases need a robust CI/CD pipeline:

1. Meet coding, documentation and sign-off standards
2. Re-create the model in a production-like environment
3. Revalidate model accuracy
4. Perform explainability checks
5. Meet all governance requirements
6. Check the quality of data artifacts
7. Test resource usage under load
8. Embed into the larger application with integration tests

Regulated industries (finance, pharmaceuticals) add manual checks; the aim is still to automate as far as possible for faster deployment, more regression testing and fewer errors.

Q6. Explain model monitoring from the perspectives of the DevOps team, the data scientist and the business. (8 marks)

Topic: Monitoring · Ch. 3, pp. 29–32

Good performance means different things to different people, so monitoring has three viewpoints.

DevOps concerns (2 marks): Is the model fast enough? Is it using sensible memory and processing time? This is traditional IT performance monitoring that DevOps already does well. Compute scalability matters, e.g., when retraining in production; deep learning needs more resources than decision trees.

Data scientist concerns (4 marks): Models degrade over time because they are models of the data they were trained on, and the real world changes. A fraud model trained six months ago misses a new fraud type; an ad model degrades as a site's audience gets younger. Retraining timing depends on how fast the world changes, required accuracy and how easy it is to deploy a better model. Degradation is detected two ways:

- **Ground truth:** the correct answer to the model's question (was this transaction really fraud?). It can arrive in seconds (ad clicks) or a month later (cardholder reports), too slow for daily monitoring.
- **Input drift:** compare recent requests against training data; distinct differences signal likely compromised performance. No waiting for ground truth is needed.

Business concerns (2 marks): Is the model delivering value? Do benefits outweigh development and deployment costs? The original KPIs should be monitored, automatically where possible. With no "dollar-o-meter", monitoring business KPIs against a sensible baseline (e.g., a rule-based model built on subject matter expertise) isolates the ML contribution.

Q7. Explain iteration and the life cycle of ML models, including retraining pitfalls and the feedback loop. (8 marks)

Topic: Iteration and Life Cycle · Ch. 3, pp. 32–34

Reasons for a new version (1 mark): performance degradation from drift, refined business objectives or KPIs, or a better model design.

Iteration (3 marks): In fast-moving settings new data arrives daily, and retraining and redeployment are automated. Even simple retraining has pitfalls:

- Does the new training data look as expected? Automated validation with predefined metrics is essential.
- Is the data complete and consistent?
- Are feature distributions similar to the previous training set? The goal is to refine, not radically change.

The new version is compared with the live model on the same development dataset. Wide variation means no automatic redeployment; manual intervention is needed. If drift is detected but little labelled data exists, data scientists must find the cause and adjust training data, e.g., remove a feature or resample rows.

Feedback loop (4 marks): Live scoring and retraining environments are kept separate, so evaluation in the retraining environment is uncertain.

- **Shadow testing:** the new model runs alongside the incumbent; the incumbent serves all requests, the new model scores them too and logs results without returning them. Results are compared statistically; it also gives SMEs visibility of future versions.
- **A/B testing:** requests are split; each is handled by one model only. Used when outcomes need user reaction (e.g., ad clicks). The fixed-horizon test waits for a predetermined sample size; peeking early is unreliable and waiting can be costly.
- **Multi-armed bandit (Bayesian):** adaptively shifts traffic away from underperforming models, reaching conclusions faster and lowering business cost.

Q8. Explain governance in MLOps: its definition, legal obligations, GDPR principles and the two categories of governance initiatives. (8 marks)

Topic: Governance · Ch. 3, pp. 34–38

Definition (1 mark): Governance is the set of controls that ensure a business delivers on its responsibilities to all stakeholders (shareholders, employees, the public, governments). These are financial, legal and ethical obligations, all underpinned by fairness.

Legal obligations (1 mark): Industry regulations long predate ML, e.g., finance rules against mismanagement and fraud, pharmaceutical rules for public health, and broader laws ensuring a level playing field on sex, race, age or religion.

Data privacy regulation (3 marks): The 2016 EU GDPR and the 2018 California CCPA protect the public from misuse of personal data, with large impact on data-dependent ML. GDPR principles for processing personal data:

1. Lawfulness, fairness and transparency
2. Purpose limitation
3. Data minimization
4. Accuracy
5. Storage limitation
6. Integrity and confidentiality (security)
7. Accountability

The EU is also planning AI-specific legislation. Beyond law, ESG indicators and public activism push businesses toward Responsible AI: ethical, transparent and accountable use of AI.

Two categories (3 marks):

- **Data governance:** ensures appropriate use and management of data — provenance, terms of collection, accuracy, and whether PII or sensitive data is present. Data lineage is hard but essential for GDPR compliance; anonymization alone may not prevent re-identification. Accidental bias is a risk, e.g., Amazon's recruiting tool that discriminated against women (scrapped in 2018).
- **Process governance:** well-defined process steps with reviews, sign-offs and documentation, so every consideration is made at the right time (no deployment before validation) and outsiders (auditors, risk managers, compliance officers) can review decisions later.

Module 1 – 6 Mark Questions

Q9. Explain the two dimensions of Responsible AI and the interplay between MLOps and Responsible AI. (6 marks)

Topic: MLOps to Mitigate Risk (MLOps for Responsible AI) · Ch. 1, pp. 9–10

Intentionality (2 marks): Models are designed and behave in line with their purpose. Data must come from compliant, unbiased sources, with collaborative checks and balances on model bias. It also includes explainability: results should be explainable by humans, not only the system's creators.

Accountability (2 marks): Central control, management and auditing of enterprise AI — no shadow IT. It means knowing which teams use what data, how, and in which models; trusting that data is reliable and collected per regulations; and traceability, i.e., finding easily where in the pipeline something went wrong.

Interplay (2 marks): ML models lack the transparency of imperative code, so proving compliance is harder. Automation moves accountability from individual contributors (e.g., pricing or loan decisions) up to data team managers or executives. Good MLOps is needed to practise Responsible AI, and Responsible AI requires MLOps strategies.

Q10. Explain why MLOps is essential for scale. (6 marks)

Topic: MLOps for Scale · Ch. 1, pp. 10–11

- **Economies of scale (1 mark):** Moving from one or a few models to tens, hundreds or thousands with positive business impact requires MLOps discipline.
- **Minimum benefits of good practice (3 marks):**
  1. Track versioning, especially of experiments in the design phase
  2. Know whether retrained models beat previous versions and promote the better ones to production
  3. Ensure at defined periods (daily, monthly) that production performance is not degrading
- **Not optional (1 mark):** Without MLOps, teams face problems of model quality and continuity, or deploy models with real negative impact, such as biased predictions that hurt the company's reputation.
- **Transparency (1 mark):** Upper management and the C-suite should see which models are in production, their business effect, and the full data pipeline from raw data to output. MLOps provides this transparency and accountability.

Q11. Differentiate between ground truth and input drift as approaches to detect model performance degradation. (6 marks)

Topic: Monitoring · Ch. 3, pp. 30–31

Models degrade because they represent the data they were trained on, and the real world changes.

Aspect	Ground truth	Input drift
Meaning	The correct answer to the question the model was asked (e.g., was the transaction actually fraudulent?)	A difference between recent requests to the model and the training data
Principle	Knowing the true answer for each prediction shows how well the model performs	A model predicts accurately only if training data reflects the real world
Speed	Fast for ad clicks (seconds); slow for fraud (phone call in minutes, cardholder report up to a month, or never)	Immediate; all needed data already exists
Waiting needed	Yes, for labels to arrive	No
Best when	Feedback comes quickly	Rapid feedback is required

Identifying drift is one of the most important parts of an adaptable MLOps strategy and brings agility to enterprise AI.

Q12. Explain shadow testing and A/B testing in the model feedback loop. (6 marks)

Topic: Iteration and Life Cycle (The Feedback Loop) · Ch. 3, pp. 33–34

Need (1 mark): Live scoring and retraining environments are separate in large enterprises, so evaluating a new version in the retraining environment is uncertain.

Shadow testing (2 marks): The new version is deployed in the live environment beside the existing model. The incumbent serves every request; the new model scores the same request, and its result is logged but not returned. After enough requests, results are compared statistically. It also gives SMEs visibility of future versions for a smoother transition.

A/B testing (2 marks): Both models are live, but requests are split; each request goes to one model only. It suits cases like ad selection, where a prediction's quality is known only after the user reacts. Its simplest form, the fixed-horizon test, must run to a predetermined sample size; peeking early is unreliable, and not stopping early can be costly.

Multi-armed bandit (1 mark): A Bayesian, adaptive alternative that shifts traffic away from underperforming models, reaching conclusions faster and reducing business cost.

Q13. Explain the challenges of iterating ML models on the edge with suitable examples. (6 marks)

Topic: Iteration and Life Cycle (Iterating on the Edge) · Ch. 3, p. 34

Challenge (1 mark): Models deployed on millions of devices (smartphones, sensors, cars) face different iteration challenges from corporate IT.

Centralized retraining – Tesla Autopilot (2 marks): Feedback from all model instances is relayed to a central point and training happens centrally. Autopilot runs in over 500,000 cars; fully retraining its roughly 50 neural networks takes 70,000 GPU hours.

Federated learning – Google GBoard (2 marks): Each smartphone retrains the keyboard model locally and sends only a summary of improvements to Google. Improvements from all devices are averaged to update the shared model.

Benefits of federated learning (1 mark): Personal data need not be collected centrally, each phone can use its improved model immediately, and overall power consumption falls.

Q14. Differentiate between data governance and process governance in MLOps. (6 marks)

Topic: Governance · Ch. 3, pp. 36–38

Aspect	Data governance	Process governance
Definition	A framework ensuring appropriate use and management of data	Well-defined processes ensuring governance is addressed at the right point in the model life cycle, with a full record kept
Focus	Data used, especially for training	Formal steps of the MLOps process and actions tied to them
Key questions / actions	Data provenance; how and under what terms it was collected; accuracy and currency; presence of PII or sensitive data	Reviews, sign-offs, capture of documentation
Aim	Lineage: where data came from and the downstream effect of changing it; GDPR compliance; avoiding accidental bias	Right decision at the right time (no deployment before validation); oversight by auditors, risk managers, compliance officers
Difficulties	Lineage is hard at feature level; anonymization may still allow re-identification; tools are in their infancy	Process knowledge is spread across teams; every team must adopt it; a heavy process gets subverted
Where common	Needed wherever personal data is used	Mostly in heavily regulated sectors such as finance

Example of data governance failure: an ML recruitment model learned bias against women because all-female schools were under-represented in upper management (Amazon, 2018).

Q15. What key questions must be answered when finding data to build ML models, including data governance constraints? (6 marks)

Topic: Model Development (Data Sources and EDA) · Ch. 3, pp. 24–25

Subject matter experts and data scientists search for input data together; this is often the most arduous part.

Questions for finding data (4 marks):

- What relevant datasets are available, and are they accurate and reliable?
- How can stakeholders access the data?
- What features can be created by combining sources?
- Will the data be available in real time?
- Must data be labelled with ground truth, or is unsupervised learning suitable, and at what cost?
- Which platform should be used?
- How will data be updated after deployment, and will using the model reduce the data's representativeness?
- How will the business KPIs be measured?

Data governance questions (2 marks):

- Can the datasets be used for this purpose, and under what terms of use?
- Is there PII that must be redacted or anonymized?
- Are there features, such as gender, that legally cannot be used here?
- Are minority populations represented well enough for equivalent performance on each group?

EDA then builds hypotheses, finds cleaning needs and guides feature selection.

Q16. Explain the role of documentation and explainability in Responsible AI during model development. (6 marks)

Topic: Model Development (Responsible AI) · Ch. 3, pp. 26–27

Verification and documentation (2 marks): Reproducing a model is not enough; DevOps must know what the model does, how to test it and the expected results. Regulated industries document how it was built and tuned, and critical models may be independently recoded. Automated model documentation helps, but some choices must be explained by hand.

Why explainability (1 mark): Performance measures do not explain how predictions are made. Knowing the "how" helps sanity-check the model, engineer better features and confirm fairness on features like sex, age or race.

Techniques (3 marks):

1. **Partial dependence plots:** the marginal impact of features on the predicted outcome
2. **Subpopulation analyses:** how the model treats specific subpopulations; the basis of many fairness analyses
3. **Individual model predictions (e.g., Shapley values):** how each feature value contributes to a specific prediction
4. **What-if analysis:** the sensitivity of a prediction to its inputs

Module 2 – 8 Mark Questions

Q17. What is a machine learning model? Explain it in theory and in practice with examples. (8 marks)

Topic: Machine Learning Model · Ch. 4, pp. 42–44

In theory (4 marks):

- A model is a projection of reality: a partial, approximate representation of some aspects of a real thing or process, chosen by what is available and useful.
- Once trained, it reduces to a mathematical formula that returns a result for given inputs, such as the probability of an event or an estimated value.
- Models rest on statistical theory; ML algorithms are the tools that build models from training data by finding a synthetic representation of it. Since data reflects the world at collection time, models predict well when the future resembles the past.

- **Generalization capacity** is the ability to predict accurately for cases unlike the input data, e.g., CycleGAN producing zebra-striped horses never seen in training.
- Examples: house price prediction from surface area, bedrooms, location and market state; disease prediction, a classification model that outputs a probability, sometimes with a confidence interval.

In practice (4 marks):

- A model is the set of parameters needed to rebuild and apply the formula. It is usually stateless and deterministic: the same inputs give the same outputs.
- It includes every transformation from raw input to the final formula and derived outputs (a class or decision). Whether ML-based or not, it is a computable function applied one row at a time.
- House price example: zip codes may be replaced by derived inputs such as average income, population density or proximity to amenities. Users still enter the zip code, so that transformation is part of the model.
- Outputs can be richer than one number: a fraud system returns a probability and flags fraud only above a fine-tuned threshold; some models output recommendations or decisions.
- These parts need not be one monolithic artifact, since they have different refresh rates and external sources.

Q18. Explain the required components of a machine learning model with suitable examples. (8 marks)

Topic: ML Required Components · Ch. 4, p. 44 (Table 4-1)

1. **Training data (2 marks):** Usually labelled with examples of what is modelled (supervised learning). Quality is vital. Example: World War II data from damaged planes that returned suffered from survivor bias, making it poor training data.
2. **Performance metric (2 marks):** What the model optimizes. A careless choice causes unintended consequences, like the cobra effect (a bounty on dead cobras led people to breed them). If 95% of data is class A, optimizing raw accuracy can yield a model that always predicts A and is 95% accurate yet useless.
3. **ML algorithm (1 mark):** Many algorithms exist with different pros and cons; choice depends on the task and priorities such as performance, stability, interpretability and computation cost.
4. **Hyperparameters (2 marks):** Configurations of the algorithm. The algorithm holds the basic formula, the learned parameters are the operations and operands for this task, and hyperparameters govern how the algorithm finds those parameters. Example: tree depth (number of splits) in a decision tree.
5. **Evaluation dataset (1 mark):** With labelled data, a dataset separate from training data measures performance on unseen data, i.e., generalization.

The number and complexity of these components, plus algorithm choice, make good MLOps challenging.

Q19. "Different ML algorithms present different MLOps challenges." Justify this statement. Also explain the role of computing power. (8 marks)

Topic: Different ML Algorithms and Different MLOps Challenges · Ch. 4, pp. 45–46 (Table 4-2)

All ML algorithms model patterns in past data; the quality and relevance of that experience decide their effectiveness. Each algorithm type has its own MLOps considerations (5 marks):

Type	Algorithm	MLOps consideration
Linear	Linear regression	Tends to overfit
Linear	Logistic regression	Tends to overfit
Tree-based	Decision tree	Unstable: small data changes can greatly change the optimal tree
Tree-based	Random forest	Predictions hard to understand (a Responsible AI concern); relatively slow to predict
Tree-based	Gradient boosting	Hard to understand; small changes in features or training set can radically change the model
Deep learning	Neural networks	Almost impossible to interpret; very slow to train; need much power and data — is a simpler model enough?

Governance and algorithm choice (1 mark): Highly regulated settings that must explain decisions (e.g., financial services) cannot use opaque models like neural networks and favour simpler techniques such as decision trees. The trade-off is often cost, not performance: simpler techniques need costlier manual feature engineering for equal performance.

Computing power (2 marks): ML development is inversely proportional to the cost of computing power. New algorithms appeared as power became affordable, from hand-computed linear regression to today's deep learning. Random forest and gradient boosting were expensive 20 years ago but lowered the cost of building models. Cheaper data helped less: most algorithms still keep all training data in memory rather than distributing it.

Q20. Explain feature engineering with its categories and techniques. Add a note on transfer learning. (8 marks)

Topic: Feature Engineering and Selection · Ch. 4, pp. 47–48

Definition (1 mark): Features are how data is presented to a model, telling it things it may not infer by itself.

Categories with examples (3 marks):

Category	Description	Example
Derivatives	Infer new information from existing information	Day of the week from a date
Enrichment	Add external information	Is this day a public holiday?

Category	Description	Example
Encoding	Present the same information differently	Weekday versus weekend
Combination	Link features together	Backlog size weighted by item complexity

Business-process example: from a date, derive the weekday or distance to the next public holiday, and consider each location's calendar. House-price example: use average income and population density instead of segmenting by zip code.

Techniques (3 marks):

- **Data enrichment services:** a market for complementary data beyond public open data.
- **Impact coding:** replace a category by the average target value for that category, at some information loss.
- **One-hot encoding:** a feature with three values (Raspberry, Blueberry, Strawberry) becomes three yes/no features.
- **Embeddings:** deep learning turns text and images into tables of numbers usable by ML algorithms.

Transfer learning (1 mark): Using knowledge gained on one problem to solve another, accelerating later tasks; popular in deep learning. Example: a model never trained on forks can still provide useful embeddings for a fork detector, because it learned similar human-made objects.

Q21. Explain experimentation in model development, including its goals, bias and variance, and how it is managed. (8 marks)

Topic: Experimentation · Ch. 4, pp. 49–51

Nature (1 mark): Experimentation runs through all of model development; important decisions are justified by experiments, ranging from full predictive models to statistical tests or charts.

Goals (2 marks):

1. Assess how useful a model can be built from the required components (Table 4-1)
2. Find the best modelling parameters (algorithms, hyperparameters, feature preprocessing)
3. Tune the bias/variance trade-off for a given training cost
4. Balance model improvement against computation cost — how good is good enough?

Bias and variance (2 marks):

- **High bias (underfitting):** misses rules learnable from data, often due to reductive, oversimplified assumptions.
- **High variance (overfitting):** treats noise as patterns, predicting every variation; complex and poor at generalizing.

Managing experiments (3 marks):

- Tools run experiments semi-automatically: you define the space of possibilities from prior knowledge and constraints (computation, budget).
- **Stratified modelling:** e.g., one demand model per store instead of one for all stores, since behaviour varies by store.
- Trying every combination is intractable, so set a time and/or computation budget and an acceptability threshold.
- Experiments may need rerunning whenever data or constraints change significantly; platforms preserve processing steps for repeatability and support version control and experimental branches to merge, discard or keep.

Q22. Explain how ML models are evaluated and compared, including choice of metrics and cross-testing. (8 marks)

Topic: Evaluating and Comparing Models · Ch. 4, pp. 51–53

Philosophy (2 marks): George Box noted that all models are wrong but some are useful. A model must be good enough to be useful, while avoiding the uncanny valley: seemingly good but catastrophic for a subset, e.g., an under-represented population. Evaluate in context against what existed before (a previous model or rules). A modest model can still save a lot, e.g., slightly better demand forecasts.

Suspicious perfect scores (1 mark): Most data has noise, so a near-perfect score suggests data leakage (target in inputs, or a feature available only after the target is known) or overfitting.

Choosing metrics (2 marks): The metric shapes the model (cobra effect). Accuracy suits balanced classification but fails for imbalanced classes: if the positive class is 5%, always predicting negative is 95% accurate and useless. No metric fits all; understand its maths trade-offs and business impact.

Cross-testing (2 marks): Evaluate on a holdout set not used in training. Data can be held out at several steps (metric evaluation, hyperparameter optimization). **k-fold cross-validation** rotates the held-out parts across several trainings, multiplying training time but showing metric stability. A holdout of the most recent records gives more realistic estimates and reveals drift between training and holdout.

Dataset split example, Fig. 4-2 (1 mark): A grey test set is held out. The rest is split three ways; each hyperparameter combination is trained on blue parts and validated on green parts. The grey set is used once, with the best combination.

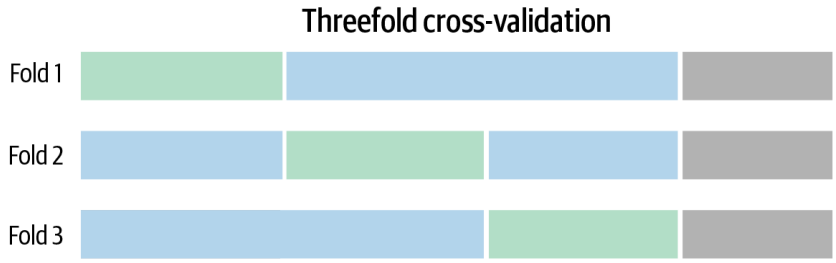


Figure 4-2. An example of dataset split for model evaluation (Source: Treveil et al., Introducing MLOps, 2020)

Q23. Explain the impact of Responsible AI on modelling, with the hiring example on fairness. (8 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 53–56

Explainability (2 marks): Some contexts need individual predictions explained, showing which features pushed a prediction. Methods: **Shapley values** (average marginal contribution of a feature value across all coalitions) and **individual conditional expectation (ICE)** (dependence of the target on features of interest). Example: a hormone level usually signals a health issue, but not for a pregnant woman. Some laws require explanations for decisions affecting people, like a loan denial.

Hiring example (4 marks): A US organization predicts worker performance to hire 10 people. The population has two hypothetical groups, Weequay and Togruta; more Weequay attend university, so Weequay applicants are more numerous and more qualified.

- **Equal opportunity:** for each applicant, all else equal, hiring must not depend on group. This yields 60% or more Weequay hires.
- **Disparate impact:** practices must not adversely affect one protected group more than another, judged on subpopulations by hiring proportions.
- The two goals are mutually exclusive: equal opportunity causes disparate impact, and forcing 40% Togruta means rejecting some Weequay predicted more qualified.
- **80% rule:** the Togruta hiring rate must be at least 80% of the Weequay rate, allowing up to 65% Weequay hires.

Implementation problems (2 marks):

- Two separate models (one per group) address over-representation but amount to disparate treatment, possibly discriminatory.
- Post-processing with a higher threshold for Weequay adjusts the trade-off but may still be disparate treatment.
- Objectives cannot be set by data scientists alone; bias is as much a business question as a technical one, and depends on context (granting privileges, negative impact like fraud rejection, or neutral like disease prediction).

Q24. Explain version management and reproducibility of ML models, listing the facets that must be documented. (8 marks)

Topic: Version Management and Reproducibility · Ch. 4, pp. 56–58

Two needs (2 marks):

1. During experimentation, go back to different branches of experiments, e.g., restore a previous project state after a dead end.
2. Let data scientists, auditors or managers replay the computations behind a deployment, possibly years later.

Versioning (1 mark): Largely solved for code via source control; platforms offer similar features for pipelines and model configuration. Merging is harder than for code, but the basic need is returning to an experiment to copy its settings.

Facets to document for reproducibility (5 marks):

1. **Assumptions:** decisions about the problem, scope and data must be explicit and logged.
2. **Randomness:** sampling and many algorithms use pseudo-random numbers; control the generator's seed.
3. **Data:** the same data must be available; versioning data can be storage-heavy and branching tools are immature.
4. **Settings:** all processing must be reproducible with the same settings.
5. **Results:** compare in-depth analyses (confusion matrices, partial dependence plots), not just text diffs.
6. **Implementation:** slightly different implementations can change close-call predictions; bulk and live API scoring may justify different implementations.
7. **Environment:** a different Python package version can change results; freeze computation environments.

Documentation can be partly automated, and an integrated platform lowers reproducibility costs. This is critical where governance and audits matter.

Module 2 – 6 Mark Questions

Q25. Explain the data exploration process carried out before training a model. (6 marks)

Topic: Data Exploration · Ch. 4, p. 46

Purpose (1 mark): Even the best algorithm is only as good as its training data. Exploration reveals issues such as incompleteness, inaccuracy and inconsistency that can make some or all data unusable.

Typical activities (4 marks):

1. Document how the data was collected and the assumptions already made

2. Examine summary statistics: each column’s domain, missing values, obvious mistakes, strange outliers (or a suspicious absence of outliers)
3. Look closely at the data’s distribution
4. Clean, fill, reshape, filter, clip and sample
5. Check correlations between columns, run statistical tests on subpopulations, fit distribution curves
6. Compare with other data or models in the literature: is usual information missing, is the distribution comparable?

Domain knowledge (1 mark): Some oddities need specialist insight. For industrial sensor data, a data scientist who is not a mechanical engineer may not know which outliers are normal for a machine.

Q26. How does feature selection impact MLOps strategy? Explain automated feature selection and feature stores. (6 marks)

Topic: Feature Engineering and Selection · Ch. 4, pp. 48–49

Trade-off (1 mark): More features may improve accuracy, give fairer finer-grained groups or compensate for missing information.

Downsides for MLOps (2 marks):

- The model becomes more expensive to compute
- More features need more inputs and more maintenance
- More features mean some loss of stability
- A large number of features can raise privacy concerns

Automated feature selection (1 mark): Heuristics estimate feature importance, e.g., correlation with the target or training a quick simple model on a subset to find the strongest predictors. Tools like Auto-sklearn or AutoML cross-reference features against the target and drop those unlikely to help.

Human choices (1 mark): Deciding whether to collect more data still needs people. Business-friendly features improve performance, ease adoption through simpler explanations and reduce modelling debt.

Feature stores (1 mark): Feature factories or stores keep features tied to business entities in a central place for reuse. They combine a slower, powerful offline part and a fast online part for real-time needs, kept consistent. They free data scientists’ time and help reduce ML’s technical debt.

Q27. Differentiate between high-bias and high-variance models. Explain stratified modelling with an example. (6 marks)

Topic: Experimentation · Ch. 4, p. 50

Aspect	High bias (underfitting)	High variance (overfitting)
Behaviour	Fails to learn rules that the training data could teach	Sees patterns in noise and tries to predict every variation
Cause	Reductive assumptions; model too simple	Model too complex for the signal
Result	Poor fit even on training data	Fits training data but generalizes poorly

Experimentation tunes this bias/variance trade-off for a given training cost (4 marks for the comparison).

Stratified modelling (2 marks): Some tools automate training one model per segment. Example: a business forecasting product demand to optimize inventory finds behaviour varies greatly between stores. Training one model per store optimizes each store better than a single model for all stores.

Q28. Explain the steps taken to cross-check model behaviour beyond raw metrics. (6 marks)

Topic: Evaluating and Comparing Models (Cross-Checking Model Behavior) · Ch. 4, p. 53

Why (1 mark): Understanding behaviour matters in proportion to the model's impact. A model that sends every patient to a doctor scores high on prevention but fails on realistic resource allocation.

Reasonable steps (3 marks):

1. Cross-check several metrics, not only the one optimized
2. Check reactions to different inputs, e.g., plot the average prediction (or probability) for values of an input and look for oddities or extreme variability
3. Split along a dimension and compare behaviour and metrics across subpopulations, e.g., is the error rate the same for males and females?

Correlation, not causality (1 mark): These analyses show how the model sees relationships, not causes. What-if analysis needs care: predictions are likely wrong for feature values, or combinations, unseen in training.

Comparison environment (1 mark): Data scientists need the full environment (tools, data) for all models to compare from every angle: for drift, same settings with different data; for performance, same data with different settings.

Q29. Explain cross-testing, k-fold cross-validation and the dataset split scheme used for model evaluation. (6 marks)

Topic: Evaluating and Comparing Models (Choosing Evaluation Metrics) · Ch. 4, p. 52 (Fig. 4-2)

Cross-testing (2 marks): To estimate generalization, the metric is computed on a holdout dataset not used for training. Data may be held out at several stages, e.g., metric evaluation and hyperparameter optimization.

k-fold cross-validation (1 mark): The held-out part is rotated so the model is trained and evaluated several times. Training time multiplies, but the spread of results shows metric stability.

Recent-records holdout (1 mark): In a simple split, holding out the most recent records mimics future use and gives more realistic estimates. It also reveals drift between training and holdout data.

Fig. 4-2 scheme (2 marks): A grey test set is set aside. The remaining data is split into three parts; each hyperparameter combination is trained three times on the blue datasets and validated on the matching green datasets. The grey set is used only once, with the best combination; the others are used with every combination.

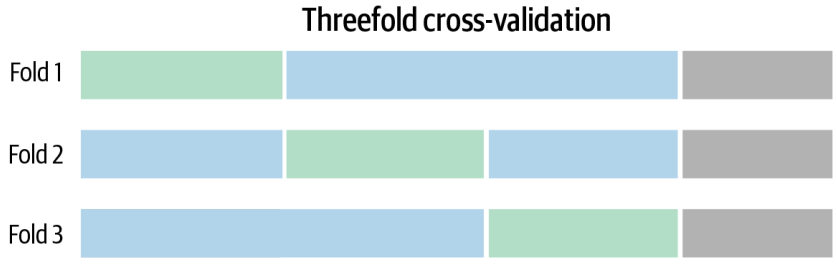


Figure 4-2. An example of dataset split for model evaluation (Source: Treveil et al., Introducing MLOps, 2020)

Q30. Explain explainability of individual predictions and why deep learning models are called black-box models. (6 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 53–54

Need (1 mark): Depending on the situation or industry, individual predictions must be explainable, showing which features pushed the prediction one way or the other. A record's prediction can differ greatly from the average.

Methods (2 marks):

- **Shapley value:** the average marginal contribution of a feature value across all possible coalitions
- **Individual conditional expectation (ICE):** shows dependence between the target function and features of interest

Example (1 mark): A high level of a certain hormone usually leads a model to predict a health issue, but for a pregnant woman the model infers no such risk. Some laws require explanations for decisions affecting people, such as recommending a loan denial.

Black-box models (2 marks): Explainability has several dimensions. Deep learning networks are fully specified by their coefficients and are conceptually simple, but the formula

is so large it cannot be grasped intuitively. Global and local tools (partial dependence plots, Shapley values) give insight but do not make the model intuitive. Rigorous, intuitive understanding requires limiting model complexity.

Q31. Explain equal opportunity, disparate impact and the 80% rule with an example. (6 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 54–55

Scenario (1 mark): A US employer uses a model to hire 10 workers. Two hypothetical groups exist, Weequay and Togruta; more Weequay attend university, so Weequay applicants are more numerous and more qualified.

Equal opportunity (2 marks): For each applicant, all else equal, hiring must not depend on group. The organization is usually not accountable for processes outside its control. Here it leads to 60% or more Weequay hires.

Disparate impact (2 marks): Employment practices must not adversely affect one protected group more than another. It is measured on subpopulations, comparing hiring proportions, usually against applicant proportions. Equal opportunity therefore creates disparate impact, while forcing 40% Togruta hires means rejecting some Weequay predicted more qualified. The two goals are mutually exclusive.

80% rule (1 mark): A legal compromise: the Togruta hiring rate must be at least 80% of the Weequay rate. Here, up to 65% Weequay hires would be acceptable. Setting these objectives is not a decision for data scientists alone.

Q32. Why must a model be evaluated in context? Why is a perfect evaluation score suspicious? (6 marks)

Topic: Evaluating and Comparing Models · Ch. 4, p. 51

"All models are wrong, but some are useful" (2 marks): This idea from statistician George E. P. Box means a model need not be perfect, only useful. Watch for the uncanny valley: a model that looks good overall but performs badly or catastrophically on a subset, such as an under-represented population.

Evaluating in context (2 marks): Compare the model with what existed before, a previous model or a rule-based process, to see the outcome of replacing it. A model with disappointing absolute performance can still improve the situation; a slightly better demand forecast may bring large cost savings.

Suspicious perfect score (2 marks): Most problems have noisy data that is hard to predict, so a perfect or near-perfect score is a warning sign of:

- **Data leakage:** the target is present in the input data, or a feature is highly correlated with the target but in practice available only once the target is known
- **Overfitting:** the model fits the training data and will not generalize well